

DAV-2 Problem Solving - I

Class started at 9:10 pm

Distribution? $\rightarrow$ Binomial distn

probability of event is given

trials (n) = 10

$$P(X < 4) = P(X=0)$$

$$+ P(X=1) +$$

$$P(X=2) + P(X=3)$$

$$P(X=k) \rightarrow \text{pmf}$$

$$P(10 < X < 70) \rightarrow \text{pdf}$$

$$P(X < 4) \rightarrow \text{cdf}$$

$$\boxed{\text{binom. cdf}} (n=10, p=0.12, k=3)$$

Test

Distribution is uniform?

0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1

$$\text{Probability of observing } T = 1 = 1/7 //$$

$$\text{not observing } T = 1 - 1/7 \\ = 6/7$$

$$= (6/7 * 6/7 * 6/7) \quad \uparrow \quad \times$$

Three times sequence

Prob of not observing T in 3 chances = $(6/7)^3$

observing T in 3 chances

$$P \text{ of observing } T = 1 - P \text{ of not observing } T = 1 - (6/7)^3 = 1 - (6/7)^3$$

$$= P_1 * P_2$$

$$1 - x$$

$$(6/7)^3 * (1 - (6/7)^3) = 0.2333 //$$

What is the probability that:

OR +

1) No T in first 3 piece sequence, AND x

2) At least one T in second 3 piece sequence

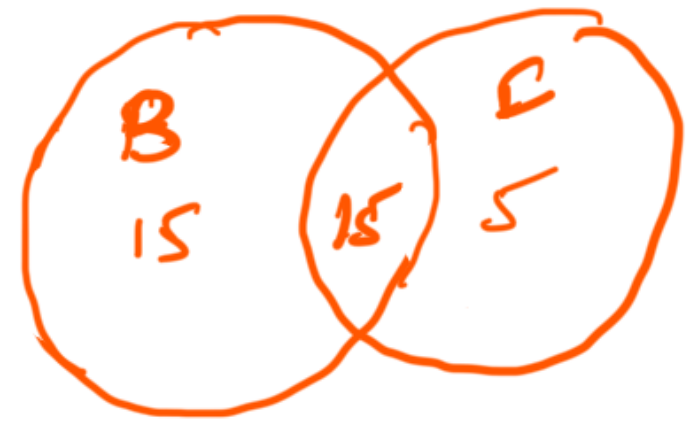
Probability

Total no of students $\rightarrow 50$

Basket ball players $\rightarrow 30$

Football " $\rightarrow 20$

Both sports $= 15$



$$\begin{aligned} P(B \cup F) &= P(B) + P(F) - P(B \cap F) \\ &= \frac{30}{50} + \frac{20}{50} - \frac{15}{50} \\ &= \frac{35}{50} = \frac{7}{10} = 0.7 \end{aligned}$$

Combination

one P, ^{one} T, ^{one} S from 10 students

$$\boxed{10C_1} \quad \boxed{9C_1} \quad \underline{8C_1}$$

$$\begin{aligned} & 10C_1 \times 9C_1 \times 8C_1 \\ &= 10 \times 9 \times 8 \\ &= 720 \end{aligned}$$

Weighted average

$$\frac{(80 \times 4 + 0 \times 0 + 90 \times 2 + 60 \times 1)}{4 + 2 + 1}$$

$$\frac{(320 + 180 + 60)}{7}$$

$$= 80$$

1/7

observing T $\rightarrow 1 - \left(\frac{6}{7}\right)^3$
 $\rightarrow 1 - \left(1 - \frac{1}{7}\right)^3$

not observing $\left(\frac{6}{7}\right)^3$

Probability (no T in first 3 And at least one T in second 3)

observing T in one = $\frac{1}{7}$ ✓

not observing T = $1 - \frac{1}{7}$ ✓
 $= \frac{6}{7}$

not obs T 3 chances = $\left(\frac{6}{7}\right)^3$ ✓

At least one t = $1 - \left(\frac{6}{7}\right)^3$ $\rightarrow 1 - \left(1 - \frac{1}{7}\right)^3$
 at least 1

when percentage, proportion or fraction is
given to you